Generative Model with Two Gamma

Data

For observations $n = 1, \ldots, N$:

- $mast_n \in \{0, 1\}$: masting indicator
- $sc_n \in \mathbb{N}_0$: seed count
- $rw_n \geq 0$: radial growth
- $DBH_n \geq 0$: DBH
- $GST_n \in \mathbb{R}$: growing season temperature

Fixed parameters:

- $\alpha > 0$: intercept for total carbon
- $\beta_{\text{growth1}} > 0$
- $\beta_{\text{growth2}} > 0$

Parameters

Carbon Allocation

$$\beta_{\text{dbh}} \in \mathbb{R}$$

$$\beta_{\text{GST}} \in \mathbb{R}$$

$$\gamma_{\text{mast}} \in (0, 1)$$

$$\gamma_{\text{nonmast}} \in (0, 1)$$

Growth

$$\sigma_{rw} > 0$$

Reproduction

$$\phi_{sc} > 0$$

$$\theta > 0$$

Transformed Parameters

Total carbon for each observation:

$$C_n = \alpha + \beta_{\text{dbh}} \cdot DBH_n + \beta_{\text{GST}} \cdot GST_n$$

Prior Distributions

$$\beta_{\text{dbh}} \sim \mathcal{N}(0, 1)$$

$$\beta_{\text{GST}} \sim \mathcal{N}(0, 1)$$

$$\gamma_{\text{mast}} \sim \text{Beta}(2, 2)$$

$$\gamma_{\text{nonmast}} \sim \text{Beta}(2, 2)$$

$$\sigma_{rw} \sim \mathcal{N}^+(0, 1)$$

$$\phi_{sc} \sim \text{Gamma}(2, 0.1)$$

$$\theta \sim \mathcal{N}^+(0, 1)$$

Model

For each observation n:

$$\gamma_n = \begin{cases} \gamma_{\text{mast}}, & \text{if } mast_n = 1 \\ \gamma_{\text{nonmast}}, & \text{if } mast_n = 0 \end{cases}$$

Growth Component

Carbon allocated to growth:

$$G_n = (1 - \gamma_n)C_n$$

Expected radial growth:

$$\mu_{rw,n} = \frac{\left(\frac{G_n}{\beta_{\text{growth1}}/100} + DBH_n^{\beta_{\text{growth2}}}\right)^{1/\beta_{\text{growth2}}} - DBH_n}{2}$$

Observation model:

$$rw_n \sim \text{LogNormal}\left(\log(\mu_{rw,n}), \sigma_{rw}\right)$$

Reproduction Component

Carbon allocated to reproduction:

$$R_n = \gamma_n C_n$$

Seed count model:

$$sc_n \sim \text{NegBinomial}_2\left(\frac{R_n}{\theta/100}, \phi_{sc}\right)$$